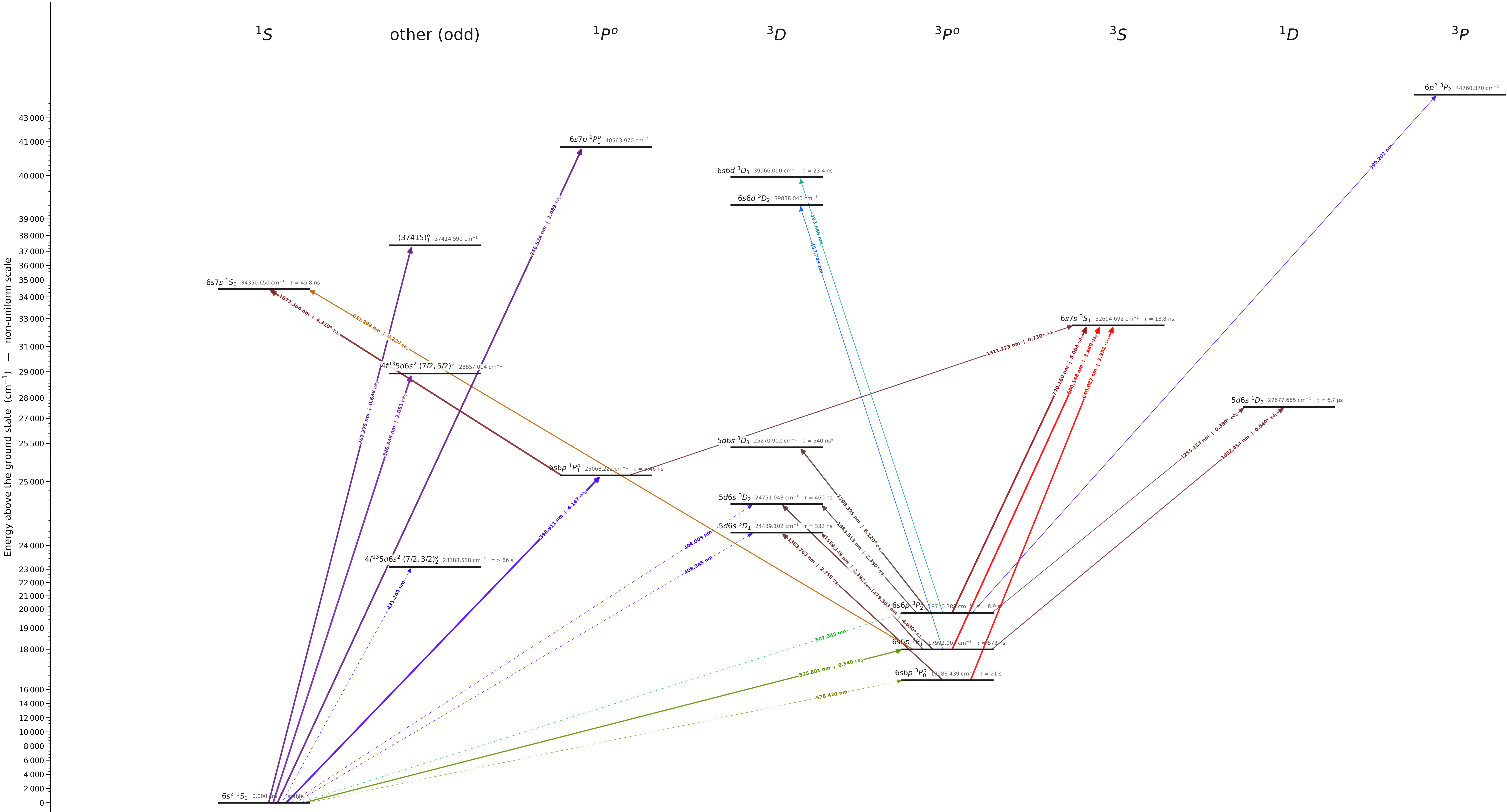
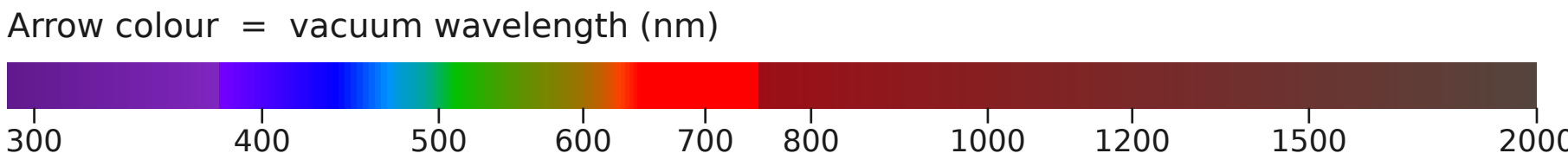


Ytterbium-171

¹⁷¹Yb (Yb I), nuclear spin *I* = 1/2. Energy levels and transitions below 2 μm: 18 levels, 26 lines.

Yb ⁺ ionisation limit 50443.20 cm⁻¹ = 6.25416 eV (198.243 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle U || e r || J' \rangle |$ in *ea*₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6s ² ¹ S ₀ – 6s7p ¹ P ₁ ^o ₁	246.5242	246.4497	1216.0772	1.489	1.000e+08	-	-	246.5 nm
6s ² ¹ S ₀ – (37415) ₁ ^o ₂	267.2754	267.1960	1121.6612	0.636	1.430e+07	-	-	267.3 nm
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 5/2) ₁ ^o ₂	346.5362	346.4370	865.1115	2.051	6.830e+07	-	-	346.5 nm
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.9108	398.7981	751.527471	4.147	1.830e+08	29.13	100	399 nm: Zeeman slower, blue MOT, imaging
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.0086	403.8945	742.0447	-	-	0.346	-	404 nm forbidden transition
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.3449	408.2296	734.1648	-	-	0.4794	-	408 nm highly forbidden
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.2489	431.1276	695.172740	-	-	2.411e-09	-	431 nm inner-shell clock transition
6s ² ¹ S ₀ – 6s6p ³ P ₂ ^o ₂	507.3452	507.2038	590.90433(4)	-	1.120e-01	1.788e-08	99.7	507 nm M2 transition: shelving into 3P2
6s ² ¹ S ₀ – 6s6p ³ P ₁ ^o ₂	555.8007	555.6464	539.388427	0.540	1.146e+06	0.1824	100	556 nm: green MOT, narrow-line cooling, imaging in tweeze...
6s ² ¹ S ₀ – 6s6p ³ P ₀ ^o ₂	578.4196	578.2592	518.295837	-	4.800e-02	7.579e-09	100	578 nm optical clock transition
6s6p ³ P ₂ ^o ₂ – 6s7s ³ S ₁	611.2976	611.1285	490.4198	0.220	4.293e+05	3.475	1.97	611 nm
6s6p ³ P ₀ ^o ₂ – 6s7s ³ S ₁	649.0866	648.9073	461.868177	1.951	9.400e+06	11.53	13	649 nm clock-state repumper / 3P0 detection
6s6p ³ P ₁ ^o ₂ – 6s7s ³ S ₁	680.1479	679.9602	440.7754	3.480	2.600e+07	11.53	36	680 nm decay channel of 3S1 during repumping
6s6p ³ P ₂ ^o ₂ – 6s7s ³ S ₁	770.1603	769.9483	389.259834	5.003	3.700e+07	11.53	50.8	770 nm 3P2 repumper
6s6p ³ P ₁ ^o ₂ – 5d6s ¹ D ₂	1032.4544	1032.1715	290.3687	0.540*	1.074e+05	0.02375	71.9	1032 nm
6s6p ¹ P ₁ ^o ₁ – 6s7s ¹ S ₀	1077.3043	1077.0093	278.2802	4.310*	3.010e+07	3.475	100	1077 nm main decay channel of 6s7s 1S0
6s6p ³ P ₂ ^o ₂ – 5d6s ³ D ₂	1255.1340	1254.7907	238.8530	0.380*	2.959e+04	0.02375	19.8	1255 nm
6s6p ¹ P ₁ ^o ₁ – 6s7s ³ S ₁	1311.2226	1310.8641	228.6358	0.730*	1.596e+05	11.53	0.21	1311 nm weak decay channel of 3S1
6s6p ³ P ₀ ^o ₂ – 5d6s ³ D ₁	1388.7630	1388.3833	215.870145	2.759	1.920e+06	0.4794	63.8	1389 nm clock-state repumper
6s6p ³ P ₁ ^o ₂ – 5d6s ³ D ₂	1479.3029	1478.8986	202.6579	4.030*	2.033e+06	0.346	93.5	1480 nm
6s6p ³ P ₁ ^o ₂ – 5d6s ³ D ₁	1539.1494	1538.7289	194.7780	2.392	1.060e+06	0.4794	35.2	1539 nm decay channel of 3D1
6s6p ³ P ₂ ^o ₂ – 5d6s ³ D ₃	1798.3949	1797.9039	166.7000	6.120*	1.864e+06	0.2947	100	1798 nm cycling transition from 3P2
6s6p ³ P ₂ ^o ₂ – 5d6s ³ D ₂	1983.5130	1982.9717	151.1422	2.390*	2.966e+05	0.346	13.6	1983 nm

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁷¹Yb hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
6s6p ³ P ₁ ^o ₁	3957.85(5)	-	F=1/2: -3957.85 F=3/2: +1978.92
6s6p ³ P ₂ ^o ₂	2681.4(26)	-	F=3/2: -4022.10 F=5/2: +2681.40
4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	1123.354(13)	-	F=3/2: -1685.03 F=5/2: +1123.35
5d6s ³ D ₁	-2038.30(11)	-	F=1/2: +2038.30 F=3/2: -1019.15
5d6s ³ D ₂	1314(3)	-	F=3/2: -1970.70 F=5/2: +1313.80
6s6p ¹ P ₁ ^o ₁	-214.06(17)	-	F=1/2: +214.06 F=3/2: -107.03
6s7s ³ S ₁	6838.41(26)	-	F=1/2: -6838.41 F=3/2: +3419.20

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	168-174	1887.64(23)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	170-174	1187.8(27)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	171-174	940.8(9)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	172-174	533.0(3)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	173-174	291.51(5)
6s ² ¹ S ₀ – 6s6p ¹ P ₁ ^o ₁	398.911	176-174	-509.25(7)
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.009	170-174	2531(5)
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.009	171-174	2019(6)
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.009	172-174	1100(4)
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.009	173-174	609(6)
6s ² ¹ S ₀ – 5d6s ³ D ₂	404.009	176-174	-1059(11)
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.345	170-174	2542(12)
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.345	171-174	2038.9(12)
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.345	172-174	1112(8)
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.345	173-174	619.4(8)
6s ² ¹ S ₀ – 5d6s ³ D ₁	408.345	176-174	-1063(6)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	171-176	-3406.79(3)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	173-176	-1797.802(26)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	174-176	-1115.787(25)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	172-176	-2296.403(25)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	170-176	-3926.431(26)
6s ² ¹ S ₀ – 4f ¹³ 5d6s ² (7/2, 3/2) ₂ ^o ₂	431.249	168-176	-5680.383(25)

Shift = ν(first isotope) – ν(second isotope).